

Brand Funds Campaign (TradeSafe Escrow Flow)

Overview

The platform uses TradeSafe as an escrow payment provider. Instead of directly paying creators, the brand first deposits money into an escrow account associated with a campaign.

The payment flow consists of four major stages:

1. Brand initiates funding for a campaign.
2. Backend creates a TradeSafe transaction and checkout link.
3. Brand completes payment on TradeSafe.
4. TradeSafe sends a webhook to notify the backend that payment has been completed.

This ensures that the backend never relies on frontend responses to determine payment success.

How I Implemented Campaign Funding

The funding process is implemented inside the `fundCampaign()` service.

Step 1. Validate Campaign

The backend first validates that:

- the campaign exists
- the campaign belongs to the logged-in brand
- the campaign is not deleted

```
Campaign.findOne({
  where:{
    id: campaignId,
    brandId,
    isDeleted:false
  }
})
```

This prevents a brand from funding another brand's campaign.

Step 2. Validate Brand TradeSafe Account

Every brand must complete TradeSafe onboarding before making payments.

The backend checks whether the brand already has a TradeSafe Token.

```
TradeSafeAccount.findOne(...)
```

If no token exists,

Brand has not completed TradeSafe onboarding.

is returned.

The TradeSafe token acts as the Brand's identity inside TradeSafe.

Step 3. Calculate Payment Amount

The backend calculates:

```
Campaign Amount
+
5% Service Charge
```

=
Total Payable

Example

Campaign Budget = ■10,000
Service Charge = ■500
Total = ■10,500

Step 4. Create TradeSafe Transaction

The backend calls createTransaction() which executes the TradeSafe GraphQL mutation transactionCreate.

TradeSafe creates:

- escrow transaction
- buyer
- seller
- first allocation

and returns TradeSafe Transaction ID.

Example
Transaction ID
TRX_7d893asd23

This ID becomes the unique identifier for the payment inside TradeSafe.

Step 5. Generate Checkout Link

After creating the transaction,
generateCheckoutLink() returns
<https://checkout.tradesafe...>

The frontend redirects the brand to this URL to complete payment.

Step 6. Store Escrow Account

The backend stores an EscrowAccount.

Important fields
campaignId
escrowReference (TradeSafe Transaction ID)
status (pending)

This record represents the campaign's escrow account inside our database.

Step 7. Store Payment Transaction

A PaymentTransaction record is also created.

It stores payment amount, provider, provider reference, payment status, checkout response.

Initially status = pending.

The database transaction is then committed.

How Campaign is Related to the Payment

Every payment belongs to exactly one campaign.

Relationship

Campaign → EscrowAccount → PaymentTransaction

Campaign

Stores campaign details.

id, title, budget, brandId

EscrowAccount

Each funded campaign creates one escrow account.

campaignId, escrowReference, status, remainingAmount

The important field is campaignId which links the escrow directly to a campaign.

PaymentTransaction

Stores every payment attempt.

campaignId, brandId, providerReference, status, amount

Again, campaignId connects the payment back to the campaign.

What is Transaction ID?

The Transaction ID is generated by TradeSafe.

Example

TRX_982734928374

It uniquely identifies a payment inside TradeSafe.

Your backend stores it as providerReference inside PaymentTransaction and escrowReference inside EscrowAccount.

Main Uses of Transaction ID

1. Generate Checkout Link

checkoutLink(transactionId)

TradeSafe needs the transaction ID to know which payment the customer is paying for.

2. Webhook Mapping

When TradeSafe sends {"id":"TRX_982734928374","state":"FUNDS_DEPOSITED"} the backend searches PaymentTransaction.findOne({ where:{ providerReference:id }}) This identifies which payment has been completed.

3. Fetch Transaction Details

Functions like getTransaction(transactionId) retrieve the latest status from TradeSafe.

4. Update Transaction

Later, updateTransaction() adds creator allocations and seller parties to the same TradeSafe transaction.

How Does the Backend Know Payment is Completed?

The backend never trusts the frontend.

Instead, it waits for TradeSafe Webhooks.

Flow

Brand → TradeSafe Checkout → Payment Successful → TradeSafe Server → Webhook →

Backend

The webhook controller receives `tradeSafeWebhookController()` which calls `processTradeSafeWebhook()`.

The payload contains `{"id":"TRX_123","state":"FUNDS_DEPOSITED"}`

The backend checks `switch(state)`

Possible states:

CREATED
FUNDS_DEPOSITED
FUNDS_RECEIVED
COMPLETED
CANCELLED
DISPUTED

FUNDS_DEPOSITED

Updates `PaymentTransaction` status = completed and `EscrowAccount` status = funded.

COMPLETED

Updates `EscrowAccount` status = fully_released.

CANCELLED

`PaymentTransaction` status = failed and `EscrowAccount` status = failed.

What Happens if the User Pays but Closes the Browser?

This is one of the biggest advantages of using webhooks.

Suppose:

Brand clicks Pay → Payment succeeds → Browser crashes / Internet disconnects / Laptop shuts down / Tab closes.

The frontend never informs your backend.

However, TradeSafe still sends the webhook because the payment was completed on its servers.

Your backend receives `FUNDS_DEPOSITED`, updates `PaymentTransaction` completed and `EscrowAccount` funded.

Therefore, the payment is recorded correctly even if the user never returns to the website.

What if the Webhook Never Arrives?

Based on the current implementation, the system depends entirely on the webhook to update payment status.

In that case:

- The payment may have actually succeeded in TradeSafe.
- `PaymentTransaction.status` will remain pending.
- `EscrowAccount.status` will remain pending.
- The campaign will appear as not funded until the webhook is received or a manual reconciliation is performed.

A common production enhancement is to add a reconciliation job that periodically calls `getTransaction(transactionId)` for pending payments and synchronizes their status with TradeSafe. This provides recovery if a webhook is delayed or temporarily fails.

Complete Funding Flow

Brand → Click Fund Campaign → Backend validates Campaign & Brand → Create TradeSafe Transaction → Receive Transaction ID → Generate Checkout Link → Save EscrowAccount → Save PaymentTransaction (Pending) → Redirect Brand to TradeSafe Checkout → Brand Pays → TradeSafe Sends Webhook → Backend Updates Payment Status → Escrow becomes Funded

This documentation accurately reflects how your current implementation works, including its strengths (server-side verification through webhooks) and its current limitation (no automatic reconciliation if a webhook is missed).